

Report of
INTERNSHIP
(Deloitte Data Analytics Internship)

Submitted
To
School Of Computer Science And Engineering
IILM University, Greater Noida, U.P.

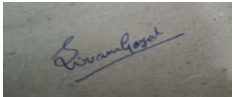
In partial fulfilment of the requirement of degree of
B.Tech CSE



By
2410030376
Shivam Goyal
3CSE2
2026-27

CANDIDATE'S DECLARATION

I, Shivam do hereby declare that the internship report titled “**Topic**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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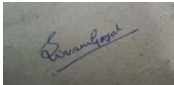
Signature

Student name: - Shivam Goyal

Roll No: - 2410030376

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

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Signature

Student Name: - Shivam Goyal

Roll No:- 2410030376

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4. PROJECT DESCRIPTION

4.1 Introduction

In today's data-driven corporate environment, organizations rely on deep analytical insights and rigorous data governance to maintain operational efficiency and internal equity. This project, based on a professional job simulation, encompasses two critical domains of data consulting: **Data Analysis** and **Forensic Technology**. The project involves transforming raw telemetry data into actionable visual insights and conducting a sensitive forensic investigation into internal human resources data to identify instances of unfair pay. By bridging data visualization with forensic classification, this project demonstrates how data can be leveraged to both optimize performance and ensure ethical compliance.

4.2 Organization Profile

Deloitte is a leading global professional services and consulting firm that specializes in audit, assurance, consulting, and tax services. The firm's Data Analytics and Forensic Technology practices are dedicated to helping clients navigate complex challenges, mitigate risks, and uncover hidden value within their data. By employing advanced technological tools and rigorous analytical frameworks, **Deloitte** assists organizations in transforming raw datasets into strategic assets and maintaining regulatory and ethical standards.

4.3 Problem Statement the project addresses two distinct but complementary client challenges:

1. **Lack of Operational Visibility:** The client generates vast amounts of telemetry data but lacks a centralized, intuitive method to explore and understand this data, leading to missed operational insights.
2. **Equity and Compliance Risks:** The client is facing concerns regarding internal pay discrepancies. They require a rigorous, evidence-based investigation to classify employee data and determine if unfair pay practices exist within the organization.

4.4 Project Objectives

The primary objectives of this project are to:

- **Analyse and process** the client's raw telemetry data to identify key trends and metrics.
- **Design and build** an interactive dashboard using Tableau to make complex telemetry data easily explorable for non-technical stakeholders.
- **Apply forensic technology principles** to structured HR/payroll data to support an internal investigation.
- **Classify and categorize** data within Microsoft Excel to highlight discrepancies and draw evidence-based conclusions regarding unfair pay.

4.5 Scope of the Project

The scope of this project is strictly defined by the parameters of the job simulation:

- **In-Scope:** Ingestion of provided client telemetry datasets, creation of a functional Tableau dashboard, logical classification of provided payroll/HR data using Excel formulas, and the formulation of final analytical conclusions.
- **Out-of-Scope:** Live database integration, real-time data streaming, deployment of the dashboard to enterprise servers, and legal/regulatory advisory services resulting from the unfair pay investigation.

4.6 Technologies and Tools Used

1. Tableau: Used for advanced data visualization, aggregating telemetry metrics, and building the interactive client-facing dashboard.

2. Microsoft Excel: Utilized for data manipulation, cleaning, and applying logical functions (e.g., IF, VLOOKUP, XLOOKUP) to classify employee records for the forensic investigation.

3. Raw Datasets (CSV/XLSX): The simulated telemetry and human resources data files provided by the client.

4.7 System Architecture

While this project does not involve building a live software application, the data architecture follows a standard analytical pipeline:

1. **Data Source Layer:** Raw telemetry files and unclassified HR spreadsheets.
2. **Processing Layer:** Microsoft Excel serves as the data transformation engine for the forensic task (adding classification columns, filtering, and sorting).
3. **Visualization Layer:** Tableau serves as the presentation engine, connecting to the processed telemetry data to render interactive charts and graphs.
4. **Insights Layer:** The final output layer where end-users interact with the dashboard and review the forensic findings.

4.8 Methodology

The project execution was divided into two distinct phases, following standard consulting workflows:

Phase 1: Data Analysis (Telemetry Dashboard)

- **Requirement Analysis:** Reviewed the client's goals to understand which telemetry metrics were most critical for their operational oversight.
- **Data Ingestion:** Imported the raw telemetry dataset into Tableau, ensuring proper data typing and relationship mapping.
- **Visualization Design:** Created individual worksheets featuring relevant charts (e.g., time-series lines, bar charts for categorical data) to highlight trends.

- **Dashboard Assembly:** Combined the worksheets into a single, cohesive dashboard, adding interactive filters to allow the client to explore the data dynamically.

Phase 2: Forensic Technology (Unfair Pay Investigation)

- **Data Familiarization:** Audited the provided HR/payroll Excel sheet to understand the existing variables (e.g., gender, tenure, role, base salary).
- **Data Classification:** Engineered new columns using Excel functions to categorize employees based on predefined criteria, isolating cohorts with identical roles and tenures.
- **Comparative Analysis:** Compared compensation across classified groups to identify statistical anomalies or direct evidence of pay disparity.
- **Conclusion Generation:** Synthesized the findings into actionable insights for the client's leadership and HR teams.

4.9 Expected Outcomes

1. A fully functional, interactive **Tableau Dashboard** that empowers the client to independently explore their telemetry data and monitor system performance.
2. A **classified and structured Excel dataset** that clearly flags anomalies and categorizes employees, providing a transparent audit trail for the forensic investigation.
3. A clearer understanding of the client's internal equity status, backed by data-driven conclusions to guide their subsequent HR policy decisions.

4.10 Certificates of Completion and Other Communication Mail Proof

Deloitte.

Shivam Goyal Data Analytics Job Simulation

Certificate of Completion
June 17th, 2026

Over the period of May 2026 to June 2026, Shivam Goyal has completed practical tasks in:

Data analysis
Forensic technology



Tina McCreery
Chief Human
Resources Officer,
Deloitte

Enrolment Verification Code YdBg8dDxM2z7jmNTp | User Verification Code 6a049389a3ac894c700c6c32 | Issued by Forge

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Data Analytics & Visualization Methodology

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- Nigrini, M. J. (2020). *Forensic Analytics: Methods and Techniques for Forensic Accounting Investigations* (2nd ed.). John Wiley & Sons. (Reference for methodologies used in the unfair pay investigation)

Project Source / Simulation

- **[Organization Name]** & Forage. (2023). *Data Analytics Virtual Experience Program*. Forage Virtual Job Simulations. Retrieved from **[Insert Link to your specific Forage/Simulation program if applicable]**

